

2. Maclaurin 급수 공식

$$\textcircled{1} \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots$$

$$\textcircled{2} \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots$$

$$\textcircled{3} \sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \frac{x^9}{9!} + \dots$$

$$\textcircled{4} \cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \frac{x^8}{8!} + \dots$$

$$\textcircled{5} \quad e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$$

$$\begin{aligned} \textcircled{6} \quad (1+x)^p &= \sum_{n=0}^{\infty} \frac{p(p-1)\cdots(p-(n-1))}{n!} x^n \\ &= \sum_{n=0}^{\infty} {}_p C_n x^n = \sum_{n=0}^{\infty} \binom{p}{n} x^n \\ &= 1 + px + \frac{p(p-1)}{2!} x^2 + \frac{p(p-1)(p-2)}{3!} x^3 + \dots \end{aligned}$$

$$\textcircled{7} \quad \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + x^4 + \dots$$

$$\textcircled{8} \quad \frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n = 1 - x + x^2 - x^3 + x^4 + \dots$$

$$\textcircled{9} \quad \ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\textcircled{10} \quad -\ln(1-x) = \sum_{n=1}^{\infty} \frac{x^n}{n} = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots$$

$$\textcircled{11} \sin^{-1}x = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \dots$$

$$= \sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2 (2n+1)} x^{2n+1}$$

$$= x + \sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)(2n+1)} x^{2n+1}$$

$$\textcircled{12} \sinh^{-1}x = x - \frac{1}{6}x^3 + \frac{3}{40}x^5 + \dots$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n (2n)!}{4^n (n!)^2 (2n+1)} x^{2n+1}$$

$$= x + \sum_{n=1}^{\infty} \frac{(-1)^n 1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)(2n+1)} x^{2n+1}$$

$$\textcircled{13} \tan x = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \dots$$

$$\textcircled{14} \tanh x = x - \frac{1}{3}x^3 + \frac{2}{15}x^5 + \dots$$

$$\textcircled{15} \sec x = 1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + \dots$$

$$\textcircled{16} \operatorname{sech} x = 1 - \frac{1}{2}x^2 + \frac{5}{24}x^4 + \dots$$

$$\textcircled{17} \tan^{-1}x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \dots$$

$$\textcircled{18} \tanh^{-1}x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)} = x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + \frac{x^9}{9} + \dots$$